# Apartment for Rent Classified Dataset

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# Problem

Rental listings are priced inconsistently because they vary across structure (beds, baths, square footage), building amenities, and location (address, city, state, latitude, longitude). Stakeholders frequently lack a transparent, scalable way to benchmark a unit's price against comparable listings in the same or different market.

**This project builds a fair-value pricing system that:**

1. standardizes all listings to a monthly baseline
2. predicts expected rent using structured, spatial, and text-derived features, and
3. quantifies mispricing with residuals and confidence bands to acknowledge listings as Deal / Fair / Premium.

### Stakeholder:

### Property managers

- Set competitive pricing that maximizes revenue while reducing vacancy risk
- Identify overpriced units that may sit, and underpriced units leaving money on the table

### Landlords

- Aim to minimize vacancy risk through competitive pricing
- Use a defensible benchmark to guide rent adjustments

### Renters

- Spot underpriced "high-value" opportunities faster
- Gain negotiation confidence with evidence-based comps and pricing bands

**Decision**: Given the structural, spatial, and amenity characteristics of a rental listing, estimate its expected market price and quantify the pricing residual (Actual – Predicted) to classify listings as underpriced, fairly priced, or overpriced.

## Why This Matters

Rental mispricing creates real market inefficiencies for both owners and renters:

- **Overpriced units** tend to sit longer on the market, increasing vacancy exposure and carrying costs.
- **Underpriced units** lease quickly but leave money on the table, resulting in avoidable revenue loss.
- **Without a transparent benchmark**, stakeholders rely on intuition or noisy comps, which weakens negotiation leverage and consistency in pricing decisions.

A data-driven fair-value framework provides a defensible reference point for setting and adjusting rent—supporting faster lease-up, healthier revenue outcomes, and more confident, evidence-based decisions across the rental lifecycle.

## Proposal Objectives

- **Build a Pricing Model:** Create a model that predicts price while standardizing for price_type (e.g., converting all rent to a Monthly baseline).
- **Feature Engineering:** Transform unstructured text (amenities/body) into standardized binary features to measure the specific impact of building assets on rent price.
- **Data Integrity:** Address data sparsity and missingness through KNN imputation to ensure room counts and structural attributes are accurately represented.
- **Define Actionable Price Bands:** Create "Deal," "Fair," and "Premium" labels based on the model's confidence intervals.
- **Create a Fair-Value Score:** Convert residual + uncertainty into a 0–100 score for fast screening and prioritization
- **Mispricing Identification:** Utilize the model's residuals (the difference between predicted and actual price) to flag units that are statistically "underpriced" (good for renters) or "overpriced" (risk for landlords).
- **Model Selection:** Compare candidate models (e.g., Random Forest vs. XGBoost) using RMSE to ensure the most reliable valuation.

## Initial Hypothesis

- **Pet Premium:** Listings with pets_allowed will have a statistically significant rent premium after controlling for bedrooms, bathrooms, square_feet, and location (latitude, longitude, cityname, state).
- **Nonlinear Size Returns:** The relationship between square_feet and price is nonlinear, with diminishing price-per-square-foot as units get larger. Nonlinear models will outperform a linear baseline in RMSE.

- **Room Count & Amenity Dominance**: Features extracted from amenities, title, and body will materially improve predictive accuracy by capturing quality signals not present in structured fields (e.g., renovated, parking, in-unit laundry).
- **Listing Friction Signals:** Metadata like has_photo and fee will systematically correlate with pricing outcomes (premium/discount or higher mispricing variance), reflecting differences in listing quality and transparency.
- **Seasonality / Time Drift:** Using time, we expect rent to show time-based drift (seasonality or macro trends). Models that include time features (month/quarter) should reduce residual bias versus models that ignore timing.

**Success Metric:**

**Primary Modeling Metric:** MAE (Mean Absolute Error) or RMSE in dollars.

**Operational Metric:** Useful separation of "Underpriced" vs "Overpriced" listings to identify ‘Market Deals’

# Related Work and Background

Rental price prediction is typically framed as a supervised regression problem. Common approaches include linear regression as an interpretable baseline, followed by more flexible models such as Random Forest and Gradient Boosting to capture nonlinear relationships and feature interactions. In housing markets, structural features such as square footage, bedroom count, and bathroom count are consistently strong predictors, while tree-based models are often used to model diminishing returns and nonlinear price effects.

A well-established finding in real estate modeling is that location variables frequently dominate structural attributes. Geographic identifiers such as city, state, or zip code often explain a substantial portion of price variance. Ignoring spatial heterogeneity can result in biased estimates and poor generalization across regions.

Another common challenge in rental datasets is heavy right skew and the presence of extreme outliers. High-end listings create long tails in the price distribution, which can distort models optimized for squared-error metrics. Best practices therefore include trimming extreme values, applying log transformations, or evaluating performance using MAE instead of RMSE when robustness is important.

These observations informed our project design. Based on EDA findings showing strong skew and spatial variability, we plan to include geographic variables, consider robust evaluation metrics, and compare interpretable baselines with tree-based models capable of modeling nonlinear effects.

# Dataset introduction and EDA summary

The Apartment for Rent Classified dataset consists of 10,000 rental listings with 22 variables describing structural attributes, textual descriptions, pricing information, and geographic coordinates. Each observation represents a single rental unit listed on an online platform. Key numerical features include price, square footage, number of bedrooms and bathrooms, and geographic coordinates. Semi-structured text fields such as amenities and body provide additional contextual information about property features.

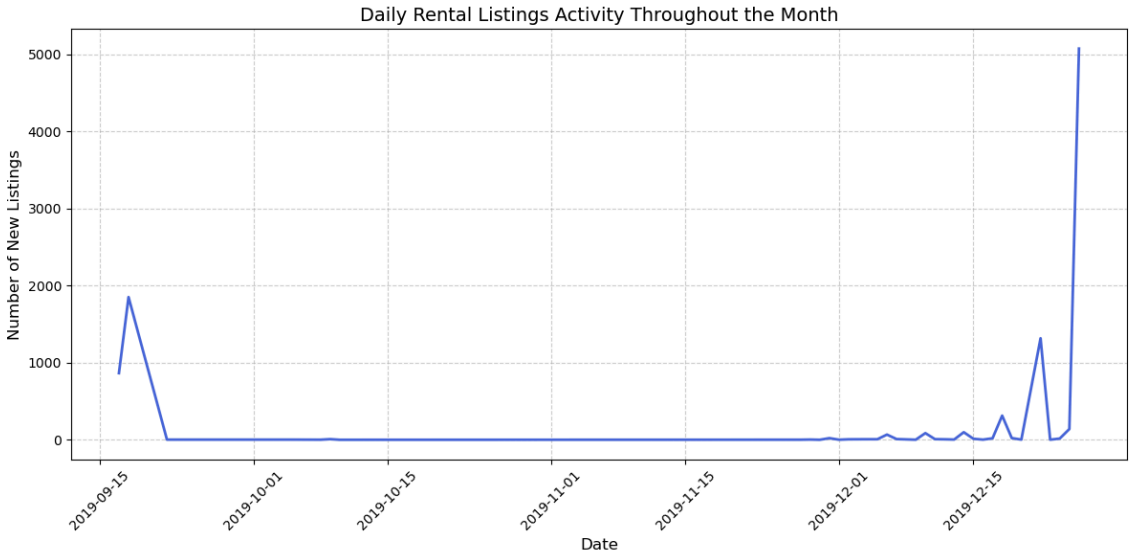
Data Definition

Category - apartment or house
Title - title of listing
Body - description of the rental unit in paragraph
Amenities - features offered in the rental unit such as parking, dishwasher, pool, etc.
Currency - all ‘USD’
Fee - all ‘No’
has_photo - either ‘thumbnail’, ‘yes’, ‘no’
pets_allowed - either 'Cats,Dogs', 'Cats', 'Dogs', empty
price and price_display - both represent the same thing, but price_display has ‘$’ with number and price is number only
Price_type - 'Monthly', 'Weekly', 'Monthly|Weekly'
Source - listing site
Time - listed date in unix timestamps

```
id                int64
category         object
title            object
body             object
amenities        object
bathrooms       float64
bedrooms        float64
currency         object
fee              object
has_photo        object
pets_allowed     object
price             int64
price_display    object
price_type       object
square_feet       int64
address          object
cityname         object
state            object
latitude        float64
longitude       float64
source           object
time              int64
dtype: object
```

```
id                  0
category            0
title               0
body                0
amenities        3549
bathrooms          34
bedrooms            7
currency            0
fee                 0
has_photo           0
pets_allowed     4163
price               0
price_display       0
price_type          0
square_feet         0
address          3327
cityname           77
state              77
latitude           10
longitude          10
source              0
time                0
```

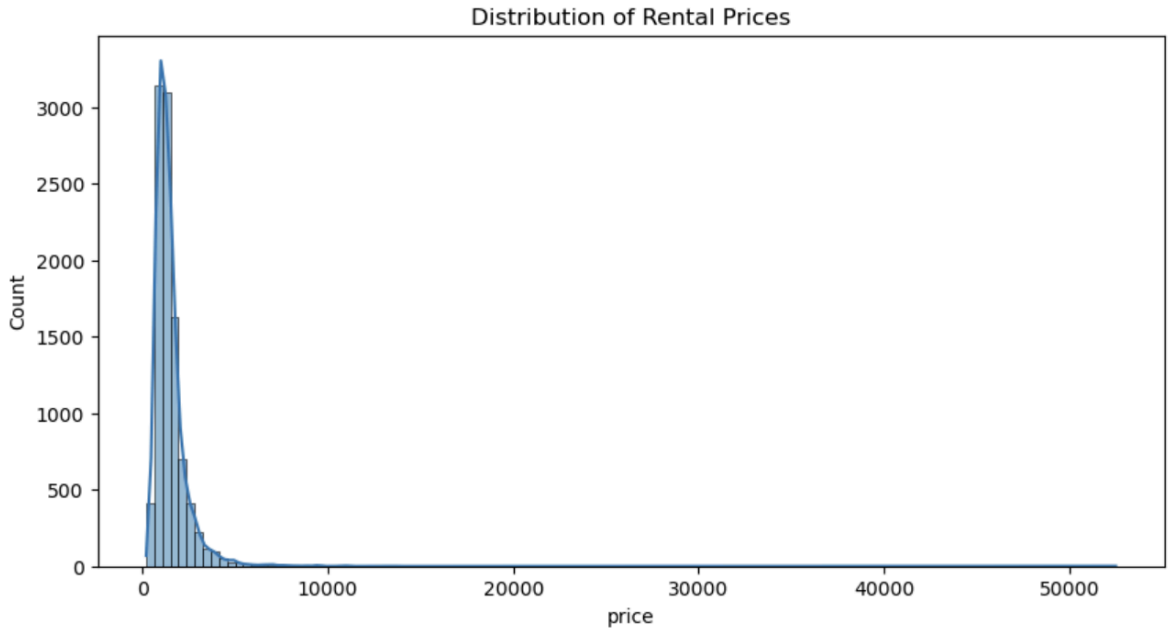
The dataset contains missing values for amenities, bathrooms, bedrooms, pets allowed, address, cityname, state, latitude and longitude.

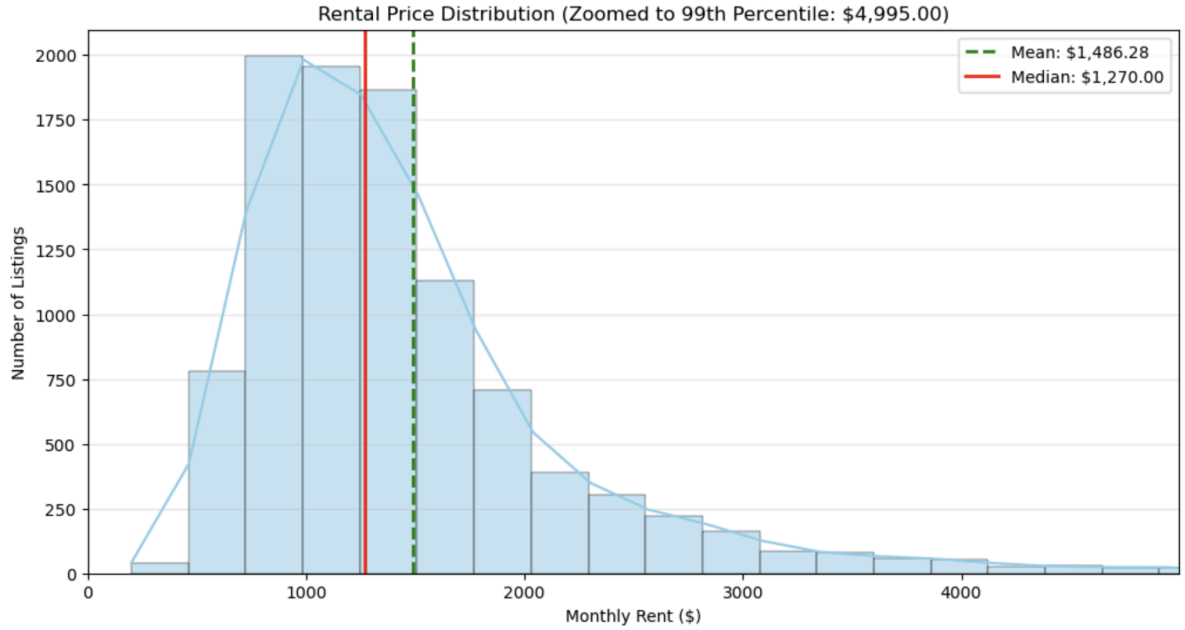



When examining daily listing activity, we identified a critical data irregularity that suggests a systemic artifact rather than natural market movement. While the dataset shows a minor cluster of activity in mid-September, the volume effectively flatlines for the remainder of the quarter. This period of inactivity is followed by an abrupt explosion of over 5,000 listings on a single day at the end of December.

Because this single-day spike contains the vast majority of our total data points, it creates a severe temporal bias. This distribution renders time-based analysis unreliable for this specific dataset. Consequently, our predictive modeling will prioritize structural and geographic features over temporal variables to ensure the results are driven by property characteristics rather than system-generated artifacts.

## Relationship Between Price and Attributes



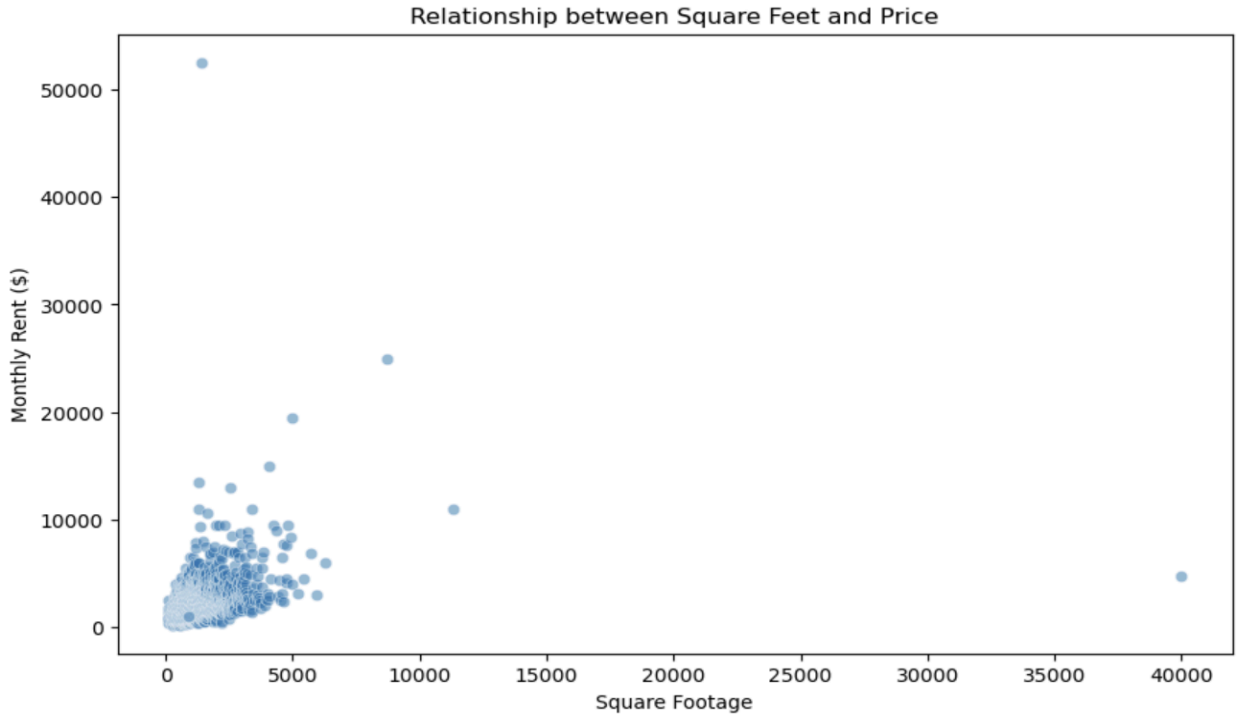

To ensure interpretability, our primary visualizations exclude outlier observations above the 99th percentile. However, the full dataset is retained for initial modeling to preserve a complete representation of the market. Rental data is naturally right-skewed because while there is a "floor" to how cheap an apartment can be, the "ceiling" for luxury penthouses is practically nonexistent. As a result, the distribution is highly concentrated between approximately $200 and $5,000 per month, with a long tail extending toward the high-end luxury market.

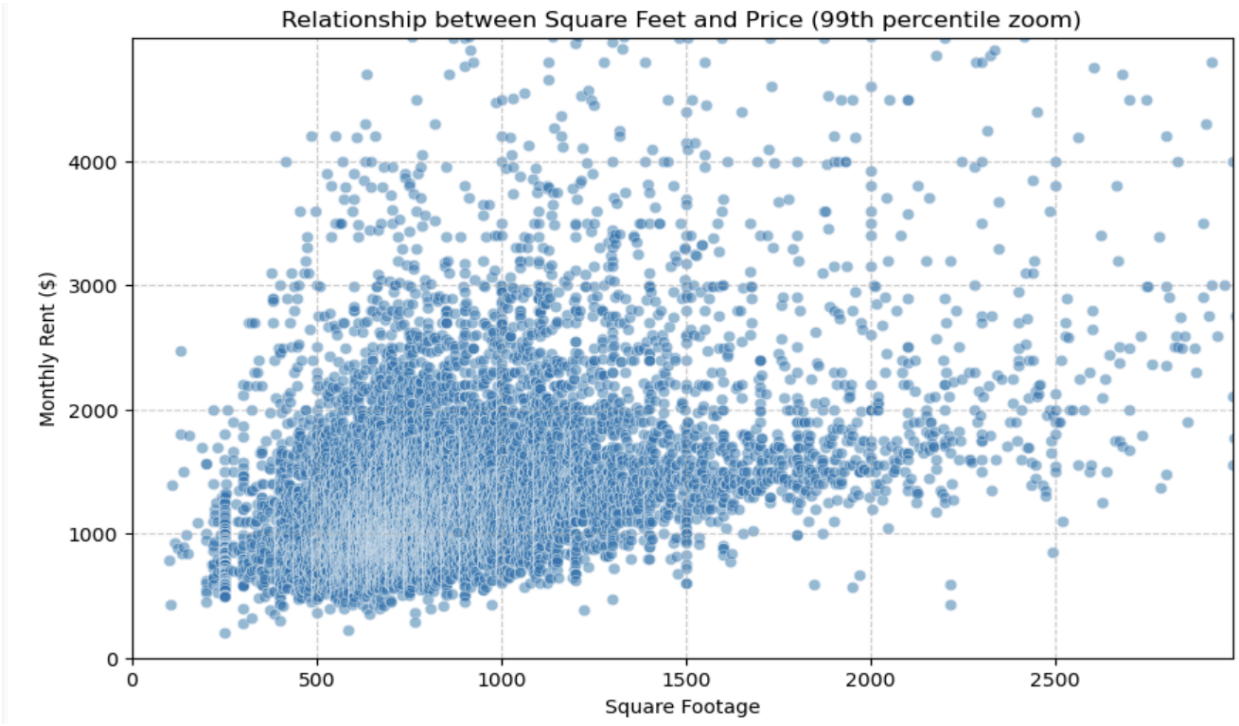
This skewness has a significant impact on how we define a "typical" price. If we rely on the mean, high-end outliers pull the average upward, making the market appear significantly more expensive than it is for the average renter. Consequently, the median serves as a fairer and more accurate measure of the simplest average price. For the average person looking at this data, the median provides a grounded reflection of the actual price center.

While the median is better for description, statistical modeling and regression models generally perform best when data follows a normal distribution. To bridge the gap between "real-world" skewness and mathematical requirements, we will explore several transformation strategies:

- **Log or Square Root Transformations**: These techniques help "pull in" the long right tail, making the distribution more symmetrical for the model.
- **Outlier Trimming**: We will distinguish between valid luxury listings and purely "noisy" data points (clerical errors) that should be trimmed to prevent model distortion.

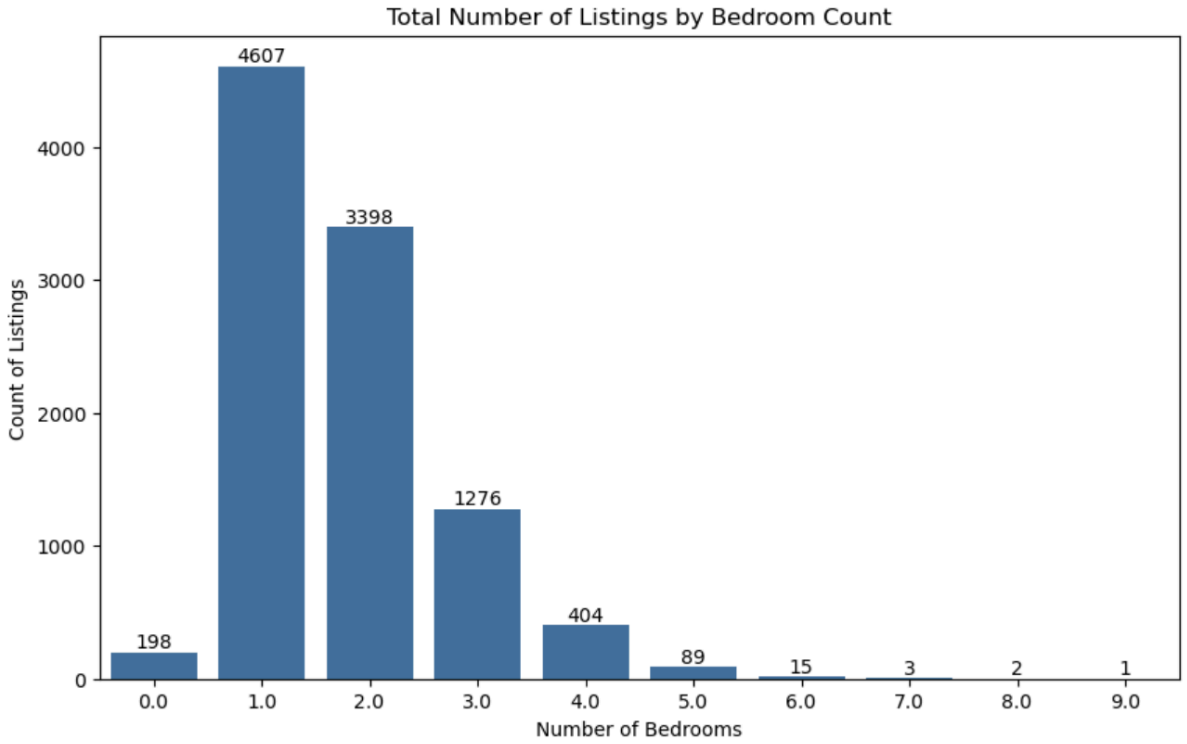
These steps ensure that our model is not just mathematically accurate, but practically useful for real-world predictions. In later stages, we will also consider robust modeling techniques specifically designed to handle such distributions without losing the predictive power of the core dataset.

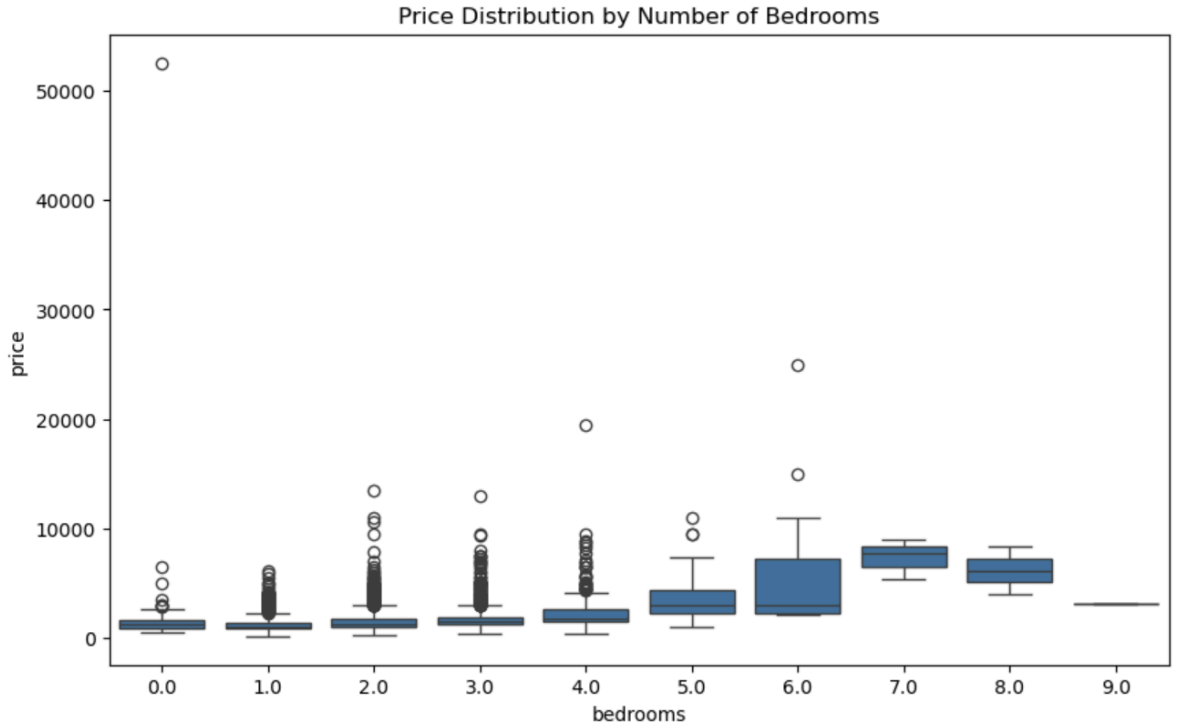


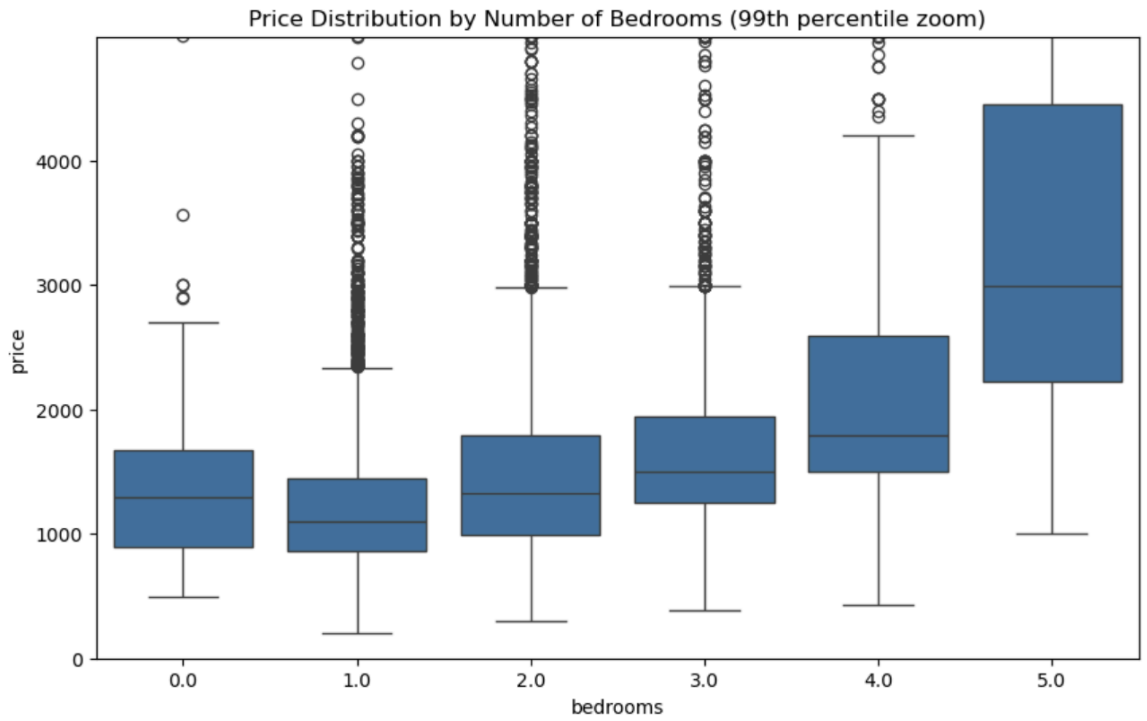


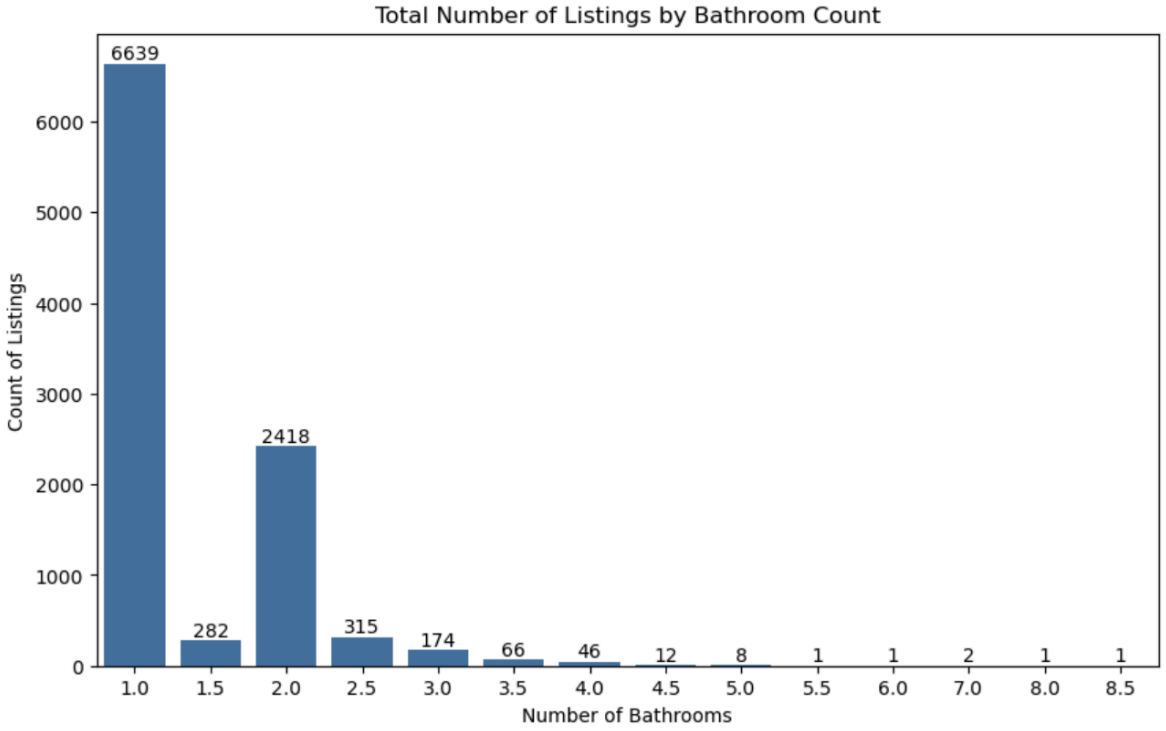

The relationship between square footage and rental cost is most reliable within the core of our dataset. The vast majority of listings are concentrated between 500 and 1,500 square feet, with corresponding rents typically falling between $1,000 and $2,500. This dense cluster represents the most liquid segment of the market, where the correlation between square footage and price remains most consistent and predictable for modeling purposes.

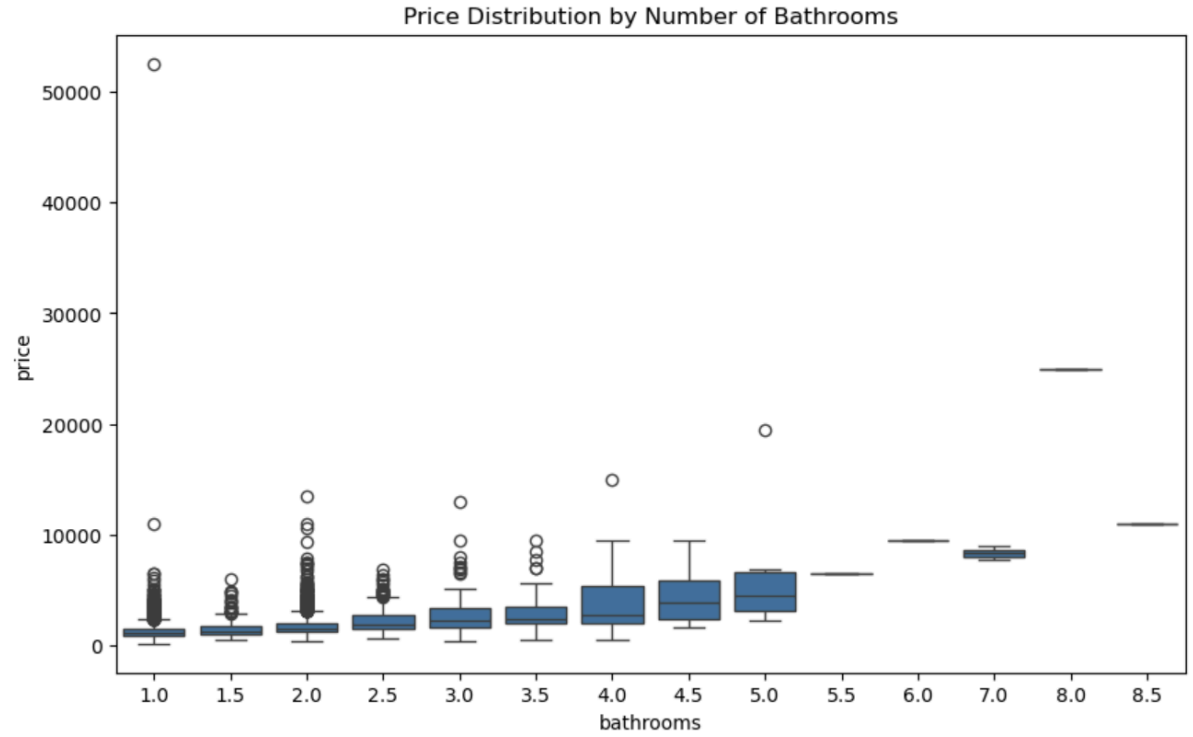
However, as properties scale into extreme dimensions, this linear relationship begins to break down not only because of the limited data on higher end rental but also the spread of the point forms a funnel shape which is detrimental for regression. A notable outlier example is a listing featuring 40,000 square feet but a relatively low rent of under $10,000. Such cases highlight a critical threshold in our data: for very large properties, square footage alone becomes a poor predictor of price. In these "long tail" scenarios, factors like specific geographic location or various amenity features likely carry more weight than pure floor area. Recognizing these limits allows us to better understand where our primary model will excel and where more complex features may be required to explain price variance.

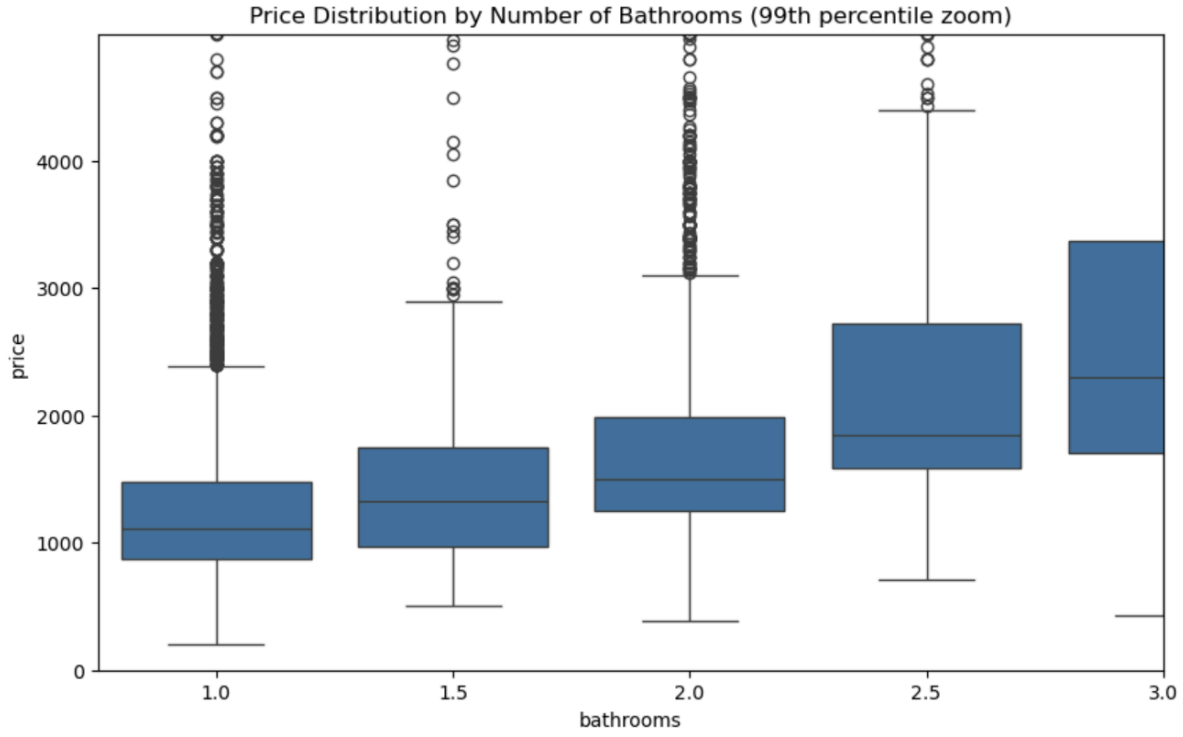

Total Number of Listings by Bedroom Count
Count of Listings
Number of Bedrooms
198
4607
3398
1276
404
89
15
3
2
1
0
1000
2000
3000
4000
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
9.0


Price Distribution by Number of Bedrooms
price
bedrooms
0
10000
20000
30000
40000
50000
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
9.0


Price Distribution by Number of Bedrooms (99th percentile zoom)
price
bedrooms
0
1000
2000
3000
4000
0.0
1.0
2.0
3.0
4.0
5.0


Total Number of Listings by Bathroom Count
Count of Listings
Number of Bathrooms
6639
282
2418
315
174
66
46
12
8
1
1
2
1
1
0
1000
2000
3000
4000
5000
6000
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
7.0
8.0
8.5




The dataset is highly concentrated within the most active segment of the rental market. Properties featuring 0 to 4 bedrooms and 1 to 3 bathrooms comprise 98% of the total data. This density is a significant strength, as it provides a statistically robust foundation for representing the most competitive and price-stable rental properties. By focusing on this core

volume, we can ensure our findings reflect broad market realities rather than niche fluctuations.

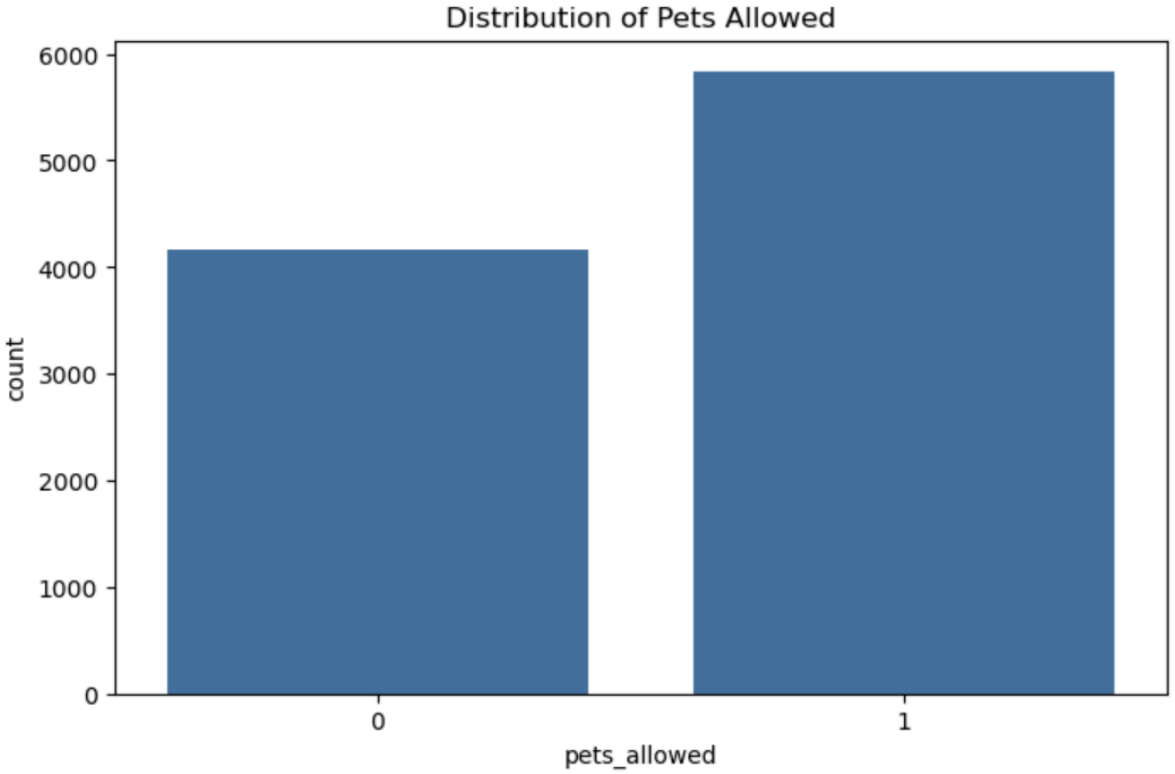
When examining price relationships, we observed a fascinating nuance in the lower end of the distribution. Interestingly, the median price for 0-bedroom units (studios) is slightly higher than that of 1-bedroom units. This likely points to a geographic premium: studios in this dataset appear concentrated in high-demand, high-cost urban centers, while 1-bedroom units are more geographically dispersed into affordable residential areas.

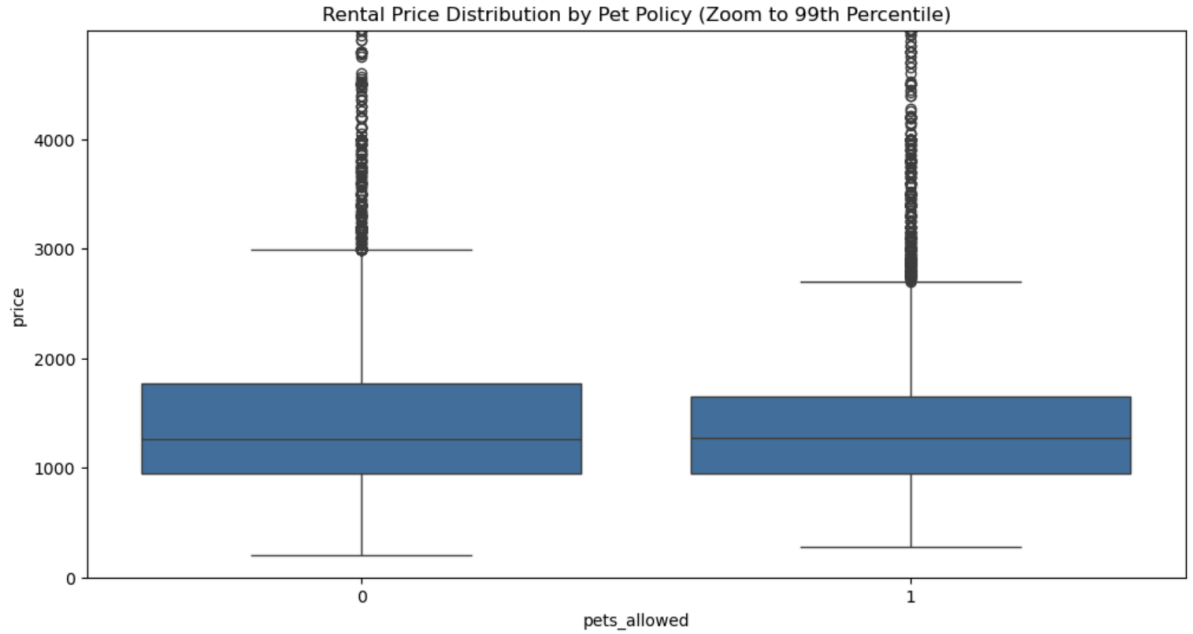
Beyond this initial threshold, the relationship between size and cost stabilizes into a highly predictable trend. Starting from the 1-bedroom mark, we see a clean, steady increase in price scaling up through 5 bedrooms. A similar linear strength is visible in bathroom counts; for example, the median price for a 2-bathroom unit is nearly double that of a 1-bathroom unit, indicating a clear linear driver for rental value.

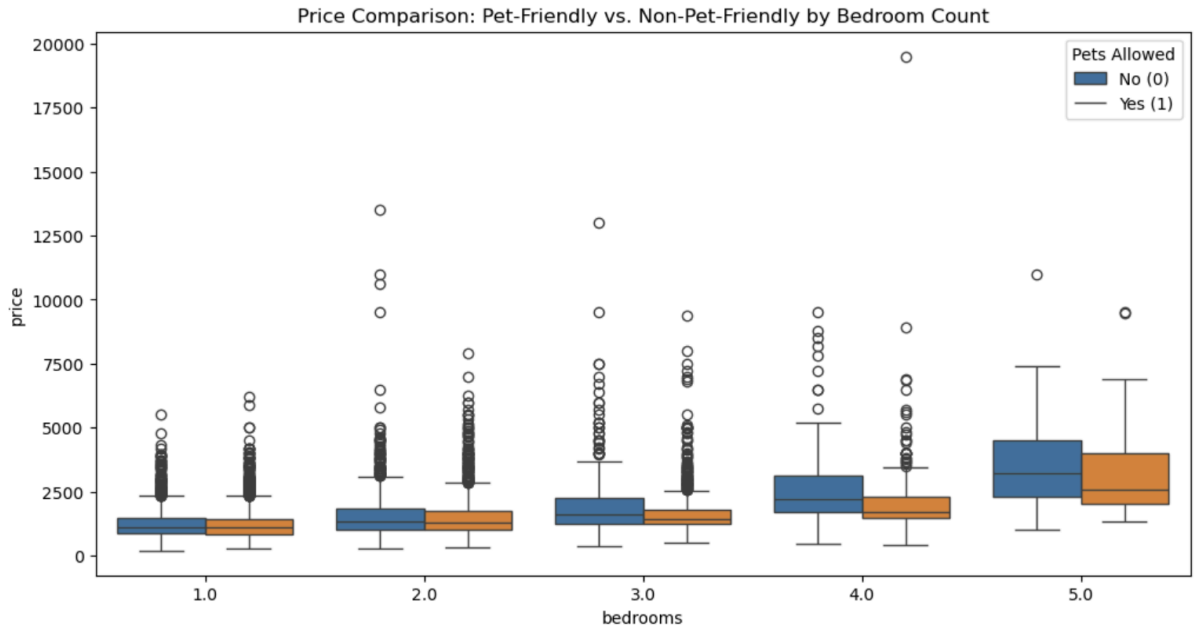
From a predictive modeling standpoint, these distributions help set clear expectations. We anticipate high accuracy for properties with 1–2 bathrooms and 0–4 bedrooms because the data is exceptionally dense and the price relationships are consistently linear.

However, we must manage expectations regarding the "long tail" of the data. Properties with 3+ bathrooms or 5+ bedrooms represent a mere 0.01% (~100/10,000) of our listings. Because the sample size for these luxury or large-scale properties is so limited, the model may struggle to generalize accurately for these edge cases.

Finally, our investigation into price outliers revealed critical manual entry errors that could otherwise degrade model performance. A notable example was a studio listed at $50,000/month for 1 bedroom, and 1 bathroom; upon cross-referencing the listing description, the actual price was found to be $500/month. To ensure our model learns from authentic market behavior rather than noise, we are implementing a rigorous cleaning phase to remove or correct these specific points before final training. This step is essential to prevent extreme outliers from skewing our coefficients and diminishing the model's overall reliability.


Distribution of Pets Allowed
6000
5000
4000
3000
2000
1000
0
count
0
1
pets_allowed

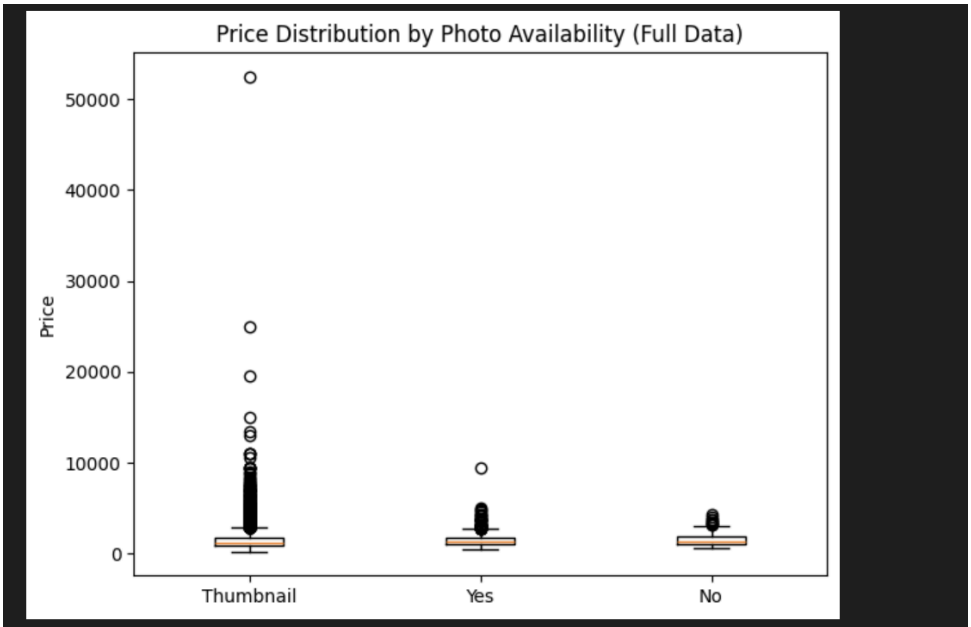

Rental Price Distribution by Pet Policy (Zoom to 99th Percentile)
4000
3000
2000
1000
0
price
0
1
pets_allowed



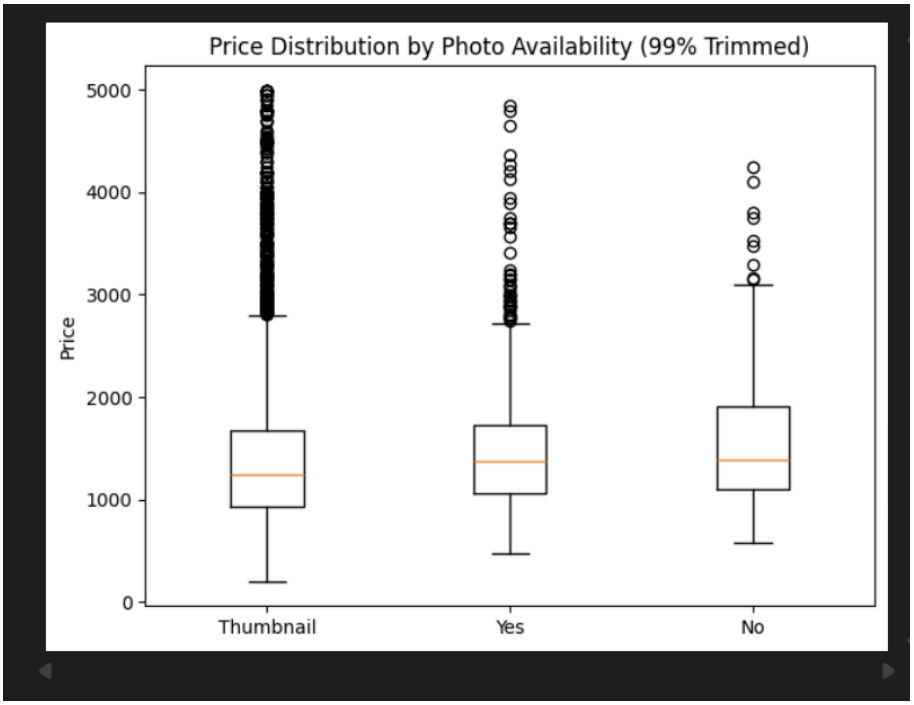

To understand the influence of secondary features on market value, we explored whether pet-friendly policies are associated with rental price premiums across various bedroom categories. The distributions for "pets allowed" and "no pets" largely overlap, suggesting that pet policy alone provides limited price differentiation.

While we observed a larger variance in price for units with more than three bedrooms, this is likely a result of data sparsity rather than a policy effect. As previously noted, instances with 0 to 3 bedrooms account for 95% of the dataset, making any observed trends in larger homes less statistically robust.

Ultimately, our analysis confirms that square footage, bedroom count, and bathroom count remain the primary structural determinants of rent. These physical characteristics exert a far stronger influence on pricing than binary amenity indicators like pet policies, which appear to have a negligible linear impact on the final rental rate.




```
99% threshold: 4995.0
Trimmed sample size: 9906
```



```
===== Group Statistics (99% trimmed) =====

Group: Thumbnail
Count: 8815
Mean: 1417.0
Median: 1250.0
Std: 714.19

Group: Yes
Count: 907
Mean: 1485.78
Median: 1379.0
Std: 627.11

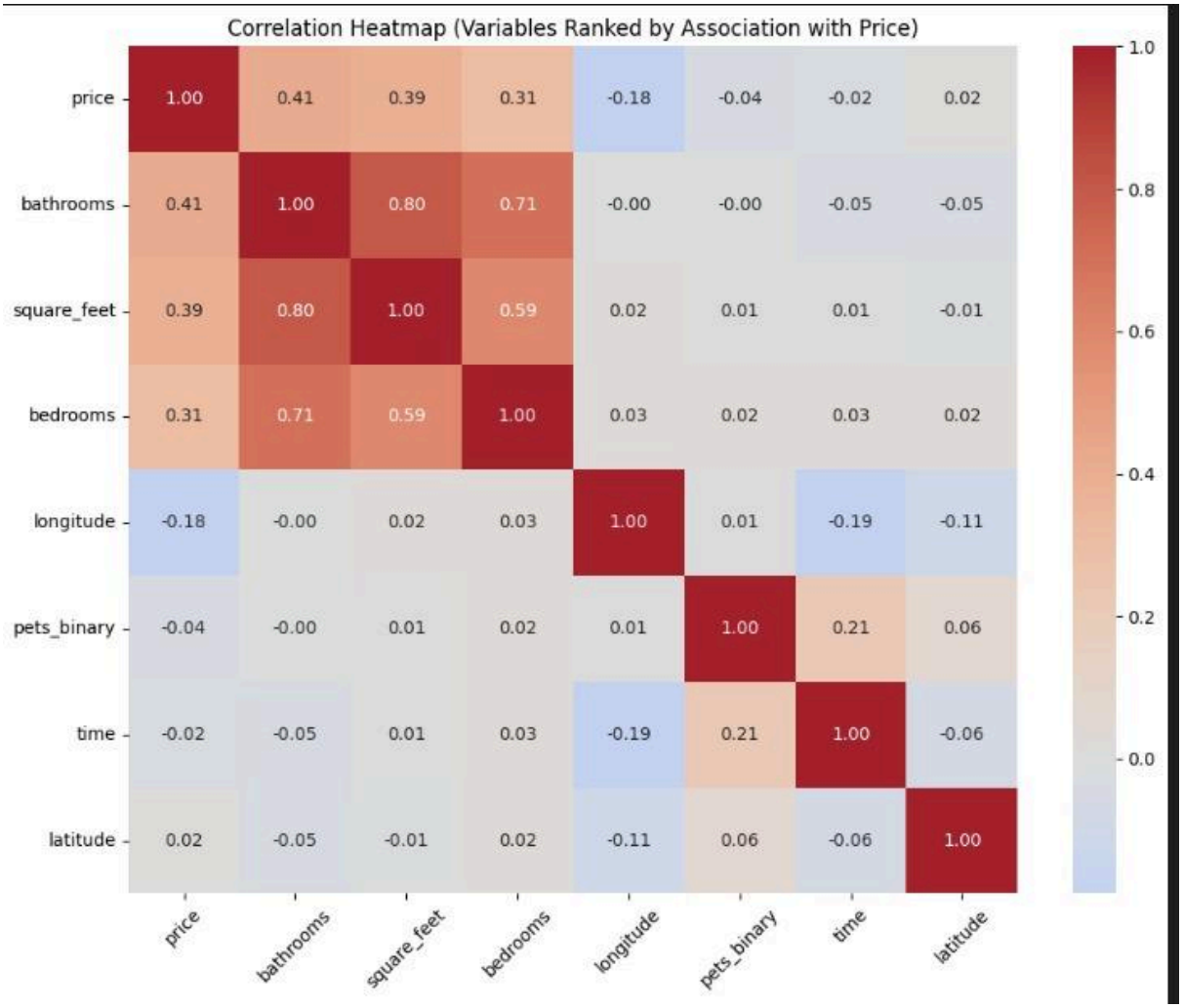
Group: No
Count: 184
Mean: 1586.52
Median: 1382.5
Std: 725.51
```

In addition to physical property characteristics, we explored whether the quality of listing presentation, specifically the availability of photos correlates with rental price. To ensure a clear comparison, this analysis was performed by grouping listings by photo status (Thumbnail, Yes, or No) and excluding extreme values above the 99th percentile.

Listings categorized with "Yes" for photo availability exhibit a higher median rent of $1,379, compared to $1,250 for those offering only a thumbnail. With only minor price differences for all three categories, it is highly probable that the observed price differences reflect the underlying quality of the properties themselves rather than the presence of a photo alone.

While this price gap is visible, it is important to note the limitations regarding the "No Photo" category. These listings represent a very small proportion of our data (n=184), which limits the statistical stability of that specific segment.

These preliminary findings indicate that while presentation-related variables correlate with pricing, their impact is likely a proxy for overall property quality. To truly isolate the effect of a photo from structural drivers like square footage or location, we will need to utilize multivariate modeling. This will allow us to determine if presentation adds independent predictive value or if it is simply a reflection of the structural attributes already being captured.

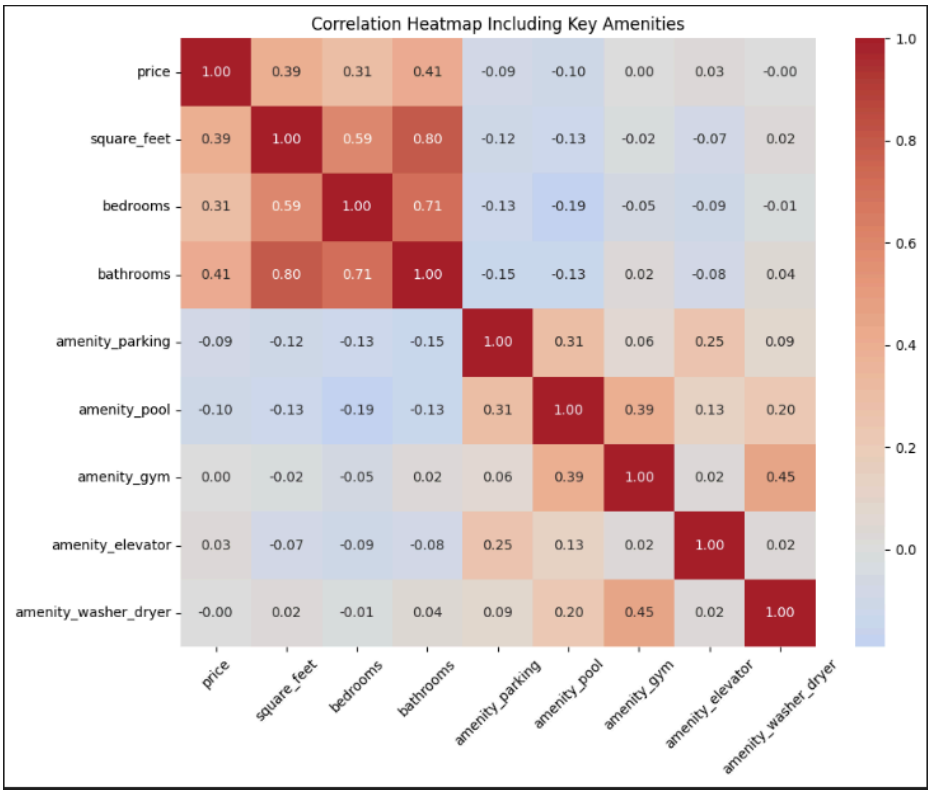


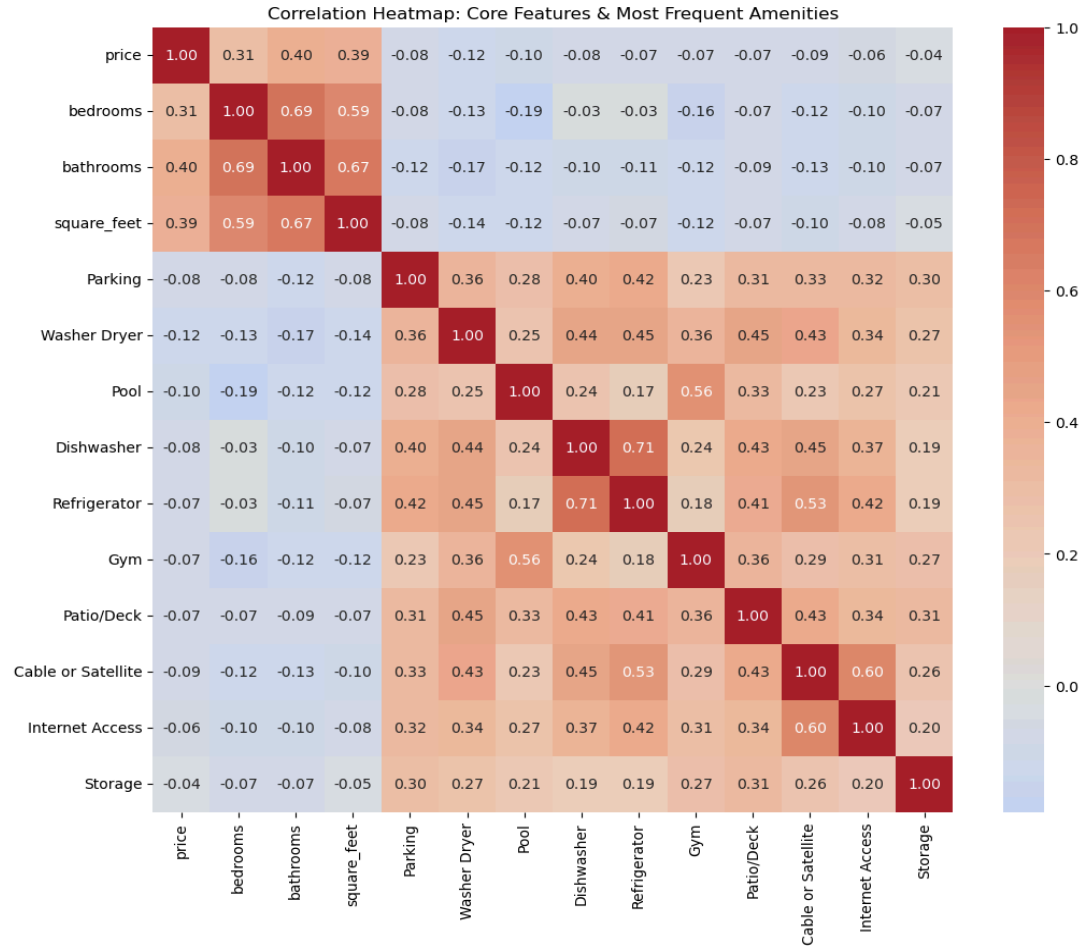

The correlation heatmap indicates that structural variables are the strongest linear predictors of rental price. Bathrooms ($r \approx 0.41$), square footage ($r \approx 0.39$), and bedrooms ($r \approx 0.31$) exhibit moderate positive correlations with price, confirming that physical property characteristics drive much of the observable price variation.

Strong multicollinearity is observed among structural features. Bathrooms and square footage show a correlation of approximately 0.80, while bedrooms and bathrooms are correlated at 0.71. This supports the use of regularized regression techniques such as Ridge to stabilize coefficient estimation.

Geographic coordinates (latitude and longitude) display weak linear correlations with price, suggesting that spatial effects may be nonlinear rather than purely linear but the problem with this comparison is that longitude or latitude alone does not give any information unless it is a location. Tree-based models may therefore provide additional value in capturing geographic heterogeneity.

The binary pet policy variable exhibits near-zero correlation with price, indicating limited direct linear impact. This changed the initial assumption that units where ‘pets_allowed’ is true will command a statistically significant price premium across all square footage levels.


Correlation Heatmap Including Key Amenities
price 1.00 0.39 0.31 0.41 -0.09 -0.10 0.00 0.03 -0.00
square_feet 0.39 1.00 0.59 0.80 -0.12 -0.13 -0.02 -0.07 0.02
bedrooms 0.31 0.59 1.00 0.71 -0.13 -0.19 -0.05 -0.09 -0.01
bathrooms 0.41 0.80 0.71 1.00 -0.15 -0.13 0.02 -0.08 0.04
amenity_parking -0.09 -0.12 -0.13 -0.15 1.00 0.31 0.06 0.25 0.09
amenity_pool -0.10 -0.13 -0.19 -0.13 0.31 1.00 0.39 0.13 0.20
amenity_gym 0.00 -0.02 -0.05 0.02 0.06 0.39 1.00 0.02 0.45
amenity_elevator 0.03 -0.07 -0.09 -0.08 0.25 0.13 0.02 1.00 0.02
amenity_washer_dryer -0.00 0.02 -0.01 0.04 0.09 0.20 0.45 0.02 1.00
price
square_feet
bedrooms
bathrooms
amenity_parking
amenity_pool
amenity_gym
amenity_elevator
amenity_washer_dryer
1.0
0.8
0.6
0.4
0.2
0.0


Correlation Heatmap: Core Features & Most Frequent Amenities
price 1.00 0.31 0.40 0.39 -0.08 -0.12 -0.10 -0.08 -0.07 -0.07 -0.07 -0.09 -0.06 -0.04
bedrooms 0.31 1.00 0.69 0.59 -0.08 -0.13 -0.19 -0.03 -0.03 -0.16 -0.07 -0.12 -0.10 -0.07
bathrooms 0.40 0.69 1.00 0.67 -0.12 -0.17 -0.12 -0.10 -0.11 -0.12 -0.09 -0.13 -0.10 -0.07
square_feet 0.39 0.59 0.67 1.00 -0.08 -0.14 -0.12 -0.07 -0.07 -0.12 -0.07 -0.10 -0.08 -0.05
Parking -0.08 -0.08 -0.12 -0.08 1.00 0.36 0.28 0.40 0.42 0.23 0.31 0.33 0.32 0.30
Washer Dryer -0.12 -0.13 -0.17 -0.14 0.36 1.00 0.25 0.44 0.45 0.36 0.45 0.43 0.34 0.27
Pool -0.10 -0.19 -0.12 -0.12 0.28 0.25 1.00 0.24 0.17 0.56 0.33 0.23 0.27 0.21
Dishwasher -0.08 -0.03 -0.10 -0.07 0.40 0.44 0.24 1.00 0.71 0.24 0.43 0.45 0.37 0.19
Refrigerator -0.07 -0.03 -0.11 -0.07 0.42 0.45 0.17 0.71 1.00 0.18 0.41 0.53 0.42 0.19
Gym -0.07 -0.16 -0.12 -0.12 0.23 0.36 0.56 0.24 0.18 1.00 0.36 0.29 0.31 0.27
Patio/Deck -0.07 -0.07 -0.09 -0.07 0.31 0.45 0.33 0.43 0.41 0.36 1.00 0.43 0.34 0.31
Cable or Satellite -0.09 -0.12 -0.13 -0.10 0.33 0.43 0.23 0.45 0.53 0.29 0.43 1.00 0.60 0.26
Internet Access -0.06 -0.10 -0.10 -0.08 0.32 0.34 0.27 0.37 0.42 0.31 0.34 0.60 1.00 0.20
Storage -0.04 -0.07 -0.07 -0.05 0.30 0.27 0.21 0.19 0.19 0.27 0.31 0.26 0.20 1.00
price
bedrooms
bathrooms
square_feet
Parking
Washer Dryer
Pool
Dishwasher
Refrigerator
Gym
Patio/Deck
Cable or Satellite
Internet Access
Storage
1.0
0.8
0.6
0.4
0.2
0.0

### Amenity Effects

Binary indicators for key amenities (parking, pool, gym, elevator, washer/dryer) exhibit weak linear correlations with rental price. This suggests that amenities contribute incrementally rather than serving as primary price determinants.

Interestingly, several amenities display slight negative correlations against both price and amenities. This does not imply that adding an amenity devalues a property; instead, it likely reflects geographic or segment effects. For instance, certain amenities may be more common in suburban developments where average rents are naturally lower than in amenity-sparse, high-priced urban centers.

Furthermore, to provide deeper context, you'll see two correlation heatmaps. While the first focuses on the core features identified by my groupmate, the second reflects an expanded feature set I created to ensure we aren't overlooking any secondary amenity correlations so there is a slight discrepancy between our correlation value due to prior data cleaning steps. We observed a couple strong correlations among amenities themselves, indicating bundled offerings where properties providing one feature often provide others such as fridge and dishwasher, pool and gym, or internet and cable/satellite.

Overall, structural variables remain substantially stronger linear predictors of rent than individual amenities. To refine our predictive strategy, we will move beyond simple correlations to explore which specific features explain the most variance to identify the strongest attribute such as PCA, Lasso regression, Tree based model, etc.

# Data Cleaning & Preprocessing Strategy

### Structural Standardization

The dataset contains three price-related fields: price, price_display, and price_type. Since 'price_display' is a formatted string representation with a '$' sign as prefix intended for user display, it was removed from the modeling dataset to avoid redundancy.

The 'price' column was retained as the primary numerical target variable. The price_type field was preserved in order to standardize rental prices to a consistent monthly baseline. Since 99.98% of observations were already priced monthly, only minimal normalization was required. The few weekly entries were converted to a monthly equivalent using a 52/12 adjustment factor. All listings were converted to a unified monthly price format to ensure comparability across observations.

The address field was removed due to substantial missingness (approximately 33%) and redundancy with geographic coordinates (latitude and longitude).

### Missing Value Treatment

The variables bedrooms and bathrooms contain a small number of missing observations (less than 0.5% of the dataset). Given the low missing rate, it will be imputed using a K-Nearest Neighbors (KNN) imputation approach based on related structural attributes such as square footage, price, city, etc. For missing city/state, we will either (1) keep an 'Unknown' category, or (2) use latitude/longitude directly as location proxies; we will not rely on external geocoding unless permitted.

This method preserves the structural consistency of listings while minimizing distortion introduced by simple mean imputation.

The variables cityname and state contain a small number of missing observations. Since geographic coordinates (latitude and longitude) are available for nearly all listings, these fields can be reconstructed using spatial proximity logic. In cases where reconstruction is not feasible, the small number of affected records may be excluded without materially impacting model performance.

## Temporal Transformation

The time variable is stored as a unix encoded timestamp. To improve interpretability and enable potential temporal analysis, this field was converted into a readable datetime format. Derived features such as listing month or year can be extracted for future modeling stages if necessary.

## Text Processing

The amenities and body text fields contain semi-structured information regarding listing features. Missing values for the 'amenities' column were standardized to empty strings to allow safe text concatenation and keyword extraction. These fields will later be transformed into structured binary indicators through controlled keyword matching.

## Data Integrity Verification

The dataset was examined for duplicate listing identifiers, structural inconsistencies, and extreme outliers in square footage values. No significant anomalies were detected in the core numerical variables, indicating that the dataset provides a stable foundation for subsequent modeling.

To ensure a reliable model, we addressed several data quality gaps through cross-checking and imputation. This included filling missing values for amenities, property details, and geoencoding like City, State using given coordinates. We also screened for duplicates and extreme outliers. With these inconsistencies resolved, the refined dataset now offers a clean and consistent basis for our analysis

### Summary

Overall, the dataset required minimal structural correction. Redundant fields were removed, price units were standardized, timestamps were converted to readable format, and text fields were cleaned for subsequent feature extraction.

# Anticipated Challenges and Constraints

### Domain Constraints

In the rental market, pricing decisions are influenced not only by structural features but also by strategic behavior. Landlords may intentionally underprice listings to secure faster occupancy or overprice them to maximize potential profit. Such behavior introduces price instability and market heterogeneity that may not be fully explained by observable property attributes.

Additionally, certain geographic areas contain very few rental listings. States such as DC, Florida, Kansas, Wyoming only have 1 city listed, and there are 684 cities with only 1 listing. The absence of comparable historical benchmarks may lead to inconsistent or poorly calibrated prices. This creates challenges for building reliable location-based pricing estimates.

From the renter's perspective, information asymmetry is a significant concern. Online listings may contain incomplete or partially inaccurate information, and rental prices may not always align with actual property quality. Therefore, any pricing model must be robust to noisy inputs and capable of providing stable and interpretable estimates rather than merely optimizing predictive accuracy.

Because this project aims to estimate fair market value and identify potential overpricing, interpretability and prediction stability are critical constraints. The model must avoid extreme volatility and produce consistent pricing signals across regions and segments.

### Data and Task Challenges

The dataset presents several structural and representational challenges.

First, missing and semi-structured information introduce preprocessing complexity. Variables such as 'pets_allowed' do not follow a simple binary format but include categories such as 'dogs', 'cats', 'dogs and cats', as well as missing or empty entries. Similarly, the amenities field contains over twenty potential features but includes approximately 3,500 missing or empty values. Some amenities (e.g., parking) appear frequently, while others (e.g., luxury) were filtered out because they were both too vague and too infrequent. Including such low-frequency, subjective terms would have added noise to the model without contributing to its overall accuracy. This uneven feature distribution complicates reliable signal extraction.

Second, the relationship between price and structural attributes is not uniformly linear. Some high-priced listings do not exhibit obvious structural differences, while certain large apartments appear relatively inexpensive on a per-square-foot basis. This suggests potential nonlinearity, diminishing returns, or unobserved quality factors influencing price formation.

Third, geographic variables introduce high cardinality and sparsity. The dataset contains over 1,500 unique cities distributed across 51 states. Many cities contain only a single listing, making direct city-level encoding prone to overfitting and unstable parameter estimation. The uneven geographic distribution further complicates generalization.

Finally, several influential factors affecting rental prices may not be fully observable in the dataset, such as interior renovation quality, neighborhood reputation, or school district effects. This latent heterogeneity introduces noise in the target variable and limits the explanatory power of purely structural models.

## Modeling Challenges

The proposed modeling framework introduces several methodological trade-offs and risks.

First, regularized linear models such as Ridge and Lasso provide interpretability and statistical stability, particularly in the presence of multicollinearity among structural features (e.g., bedrooms, bathrooms, and square footage). However, these models assume primarily linear relationships. Given EDA evidence suggesting nonlinear effects and diminishing returns to size, linear models may underfit complex pricing patterns.

Second, tree-based models such as Random Forest and XGBoost are capable of capturing nonlinear interactions and heterogeneous geographic effects. However, these flexible models introduce a significant risk of overfitting, particularly in the presence of high-cardinality geographic variables and sparse amenities features. Rare cities or infrequent amenities may be memorized rather than generalized, reducing robustness.

Third, the bias–variance trade-off becomes central to model selection. Simpler models may provide stable and interpretable estimates but fail to capture structural complexity, while more complex models may achieve higher predictive accuracy at the cost of interpretability and stability.

Finally, because overpricing classification is derived from residuals, prediction errors near the decision threshold may lead to unstable classification outcomes. Ensuring that residual-based signals are robust and not driven by noise is an additional modeling consideration.

# Methodology Direction

Our modeling strategy is designed to align with the stakeholder decision of estimating fair market rent and identifying potential overpricing in rental listings. Rather than pursuing purely predictive optimization, the goal is to build a stable and interpretable benchmark model that reflects structural rental determinants.

### Baseline Model

We will begin with a regularized linear regression model (Ridge regression) as our baseline. This choice is motivated by several factors:

1. Rental price exhibits a strong linear relationship with structural variables such as square footage, bedrooms, and bathrooms, as observed in the EDA.

2. High-cardinality categorical features (e.g., geographic variables) introduce multicollinearity and instability risks, which regularization helps mitigate.

3. Interpretability is important for stakeholders who may want to understand how features influence price estimation.

The baseline will include structural variables (bedrooms, bathrooms, square footage), geographic indicators (state-level encoding), and selected  amenity features.

### Geographic Strategy

With over 1,500 cities represented in the dataset, direct dummy encoding presents significant risks of overfitting and data sparsity. To mitigate these 'high-cardinality' issues, geographic data will be aggregated at the state level to enhance model stability and generalization. Additionally, latitude and longitude will be integrated as continuous spatial proxies; this allows the model to capture location effects without the dimensionality explosion associated with categorical variables.

### Evaluation Metrics

Due to the strong right skew in rental prices, performance will be evaluated using both RMSE and MAE. MAE will serve as the primary metric because:

- It is less sensitive to extreme outliers.
- It provides interpretable error magnitude in dollar terms.

In addition to overall error, we will examine residual distributions to assess model stability across bedroom segments and geographic groups. This allows us to verify that the model remains robust across different market environments.

### Model Comparison

To evaluate whether nonlinear modeling provides additional value, we will compare the baseline against tree-based models such as Random Forest or Gradient Boosting. However, improvements will be weighed against interpretability and stability concerns.

# Analytical Contribution

The analytical contribution of this project focuses on improving the stability and reliability of geographic price modeling.

Although linear geographic correlation appears weak, spatial heterogeneity may still influence rental price. We test whether refined geographic representation improves predictive stability. However, the dataset contains over 1,500 distinct cities, many of which appear only once or a few times. Direct city-level encoding introduces high sparsity and overfitting risk, potentially reducing model generalization.

Our contribution tests whether a refined geographic representation improves predictive stability compared to a naïve baseline.

### Baseline Model

The baseline model includes:

- Structural features (bedrooms, bathrooms, square footage)
- State-level geographic encoding

- Cleaned amenity indicators

This model provides a stable and interpretable benchmark.

## Contribution Model

The contribution model will incorporate latitude and longitude as continuous spatial predictors to capture fine-grained geographic variation while maintaining dimensional stability. As a robustness check, we will also test grouped geographic clusters based on regional proximity. The goal is to capture local price variation without introducing excessive dimensionality.

## Evaluation Strategy

The contribution will be evaluated using:

1. Change in MAE (primary metric)
2. Change in RMSE
3. Residual stability across geographic segments
4. Overfitting gap between training and testing performance

An ablation-style comparison (Baseline vs. Baseline + Geographic Refinement) will determine whether the added complexity produces meaningful and stable improvements. Residual-based classification will be defined using standardized residual thresholds (e.g., ±1 standard deviation) to ensure consistent pricing signals. This ensures that classification is based on relative pricing deviation rather than absolute dollar difference, improving comparability across regions.

# Milestone Plan

| Phase | Goal | Output | Owner(s) | Target date |
|---|---|---|---|---|
| Baseline pipeline | Complete first end-to-end modeling workflow (data - features - model – evaluation) | Cleaned dataset + baseline model (OLS & Ridge) + MAE/RMSE + diagnostic plots + summary notes | Xinyu Jing, Brandon Lu | March 3 |
| Contribution design | Define main contribution + ablation plan | Clear contribution statement + feature set comparison table + ablation experiment plan | Xinyu Jing, Brandon Lu, Wing Yan Kwan | March 16 |
| Improved evaluate | Implement improved models and conduct robustness analysis | Slice results + robustness result | Wing Yan Kwan, Tri Nguyen | March 23 |
| Final Writing + slides | Finalize narrative and prepare presentation materials | Final Report（PDF or Jupyter notebook) + deck | all | March 29 |